Programming Fundamentals — Assignment

Lab 05: File Handling in C++: Opening, Reading, Writing and Error Handling

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Date: 11-10-2026

Github Repo: https://github.com/aaliyann-exe/programming-fundamentals-lab

Task 1: Class Representative Record.

Code:

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    string marks, rollNo;

    ofstream file("students_list.txt");

    for (int i = 1; i <= 5; i++) {

        cout << "Enter roll number of student " << i << ": ";
        cin >> rollNo;
        cout << "Enter marks of student#" << rollNo << ": ";
        cin >> marks;

        file << rollNo << " " << marks << endl;

    }

    file.close();
    cout << "Student list saved to students_list.txt" << endl;

    return 0;

}
```

Output:

```
Enter roll number of student 1: 01
Enter marks of student#01: 20
Enter roll number of student 2: 02
Enter marks of student#02: 50
Enter roll number of student 3: 03
Enter marks of student#03: 67
Enter roll number of student 4: 04
Enter marks of student#04: 76
Enter roll number of student 5: 05
Enter marks of student#05: 98
Student list saved to students_list.txt
```

Task 2: Display the Saved List.

Code:

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    ifstream file("students_list.txt");

    if(!file) {

        cout << "No file found.";
        return 1;

    }

    string rollNo, marks;
    int count = 0;

    cout << "Student\tRoll#\tMarks" << endl;

    while (file >> rollNo >> marks) {

        count++;
        cout << count << "\t" << rollNo << "\t" << marks << endl;

    }

    file.close();
    cout << "Total students: " << count << endl;

    return 0;

}
```

Output:

```
Student Roll#   Marks
1       01      20
2       02      50
3       03      67
4       04      76
5       05      98
Total students: 5
```

Task 3: Creating a File and Writing Student Information.

Code:

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    ofstream file;
    file.open("student_info.txt");

    string name, rollNo, department;

    cout << "Enter name: ";
    getline(cin, name);
    cout << "Enter roll number: ";
    getline(cin, rollNo);
    cout << "Enter department: ";
    getline(cin, department);

    file << name << endl;
    file << rollNo << endl;
    file << department << endl;

    file.close();
    cout << "Student information saved." << endl;

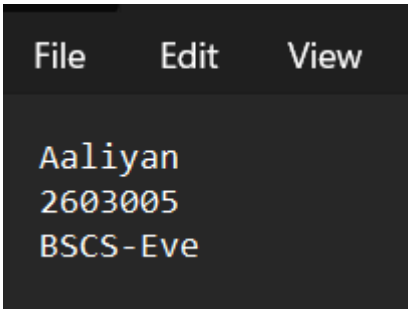
    return 0;

}
```

Output:

```
Enter name: aaliyan
Enter roll number: 2603005
Enter department: BSCS-Eve
Student information saved.
```

File:



Task 4: Reading and Displaying File Content.

Code:

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    ifstream file("student_info.txt");

    if(!file) {

        cout << "File not found.";
        return 1;

    }

    string line;

    cout << "Content of student_info.txt:" << endl;

    while (getline(file, line)) {

        cout << line << endl;

    }

    file.close();

    return 0;

}
```

Output:

```
Content of student_info.txt:
aaliyan
2603005
BSCS-Eve
```

Task 5: Appending Data to a File.

Code:

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    ofstream outFile("student_info.txt", ios::app);

    if (!outFile.is_open()) {

        cout << "Error: could not open student_info.txt" << endl;
        return 1;

    }

    string phone, email;

    cout << "Enter phone number: ";
    getline(cin, phone);
    cout << "Enter email: ";
    getline(cin, email);

    outFile << phone << endl;
    outFile << email << endl;

    outFile.close();
    cout << "Additional information appended successfully." << endl;

    ifstream inFile("student_info.txt");

    if (!inFile.is_open()) {

        cout << "File not found." << endl;
        return 1;

    }

    string line;

    cout << "\nContent of student_info.txt:" << endl;

    while (getline(inFile, line)) {
        cout << line << endl;
    }

    inFile.close();

    return 0;

}
```

Output:

```
Enter phone number: 03185949460
Enter email: aaliyan@example.com
Additional information appended successfully.

Content of student_info.txt:
aaliyan
2603005
BSCS-Eve
03185949460
aaliyan@example.com
```

Task 6: File Truncation (Overwriting Content).

Code:

```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    ofstream outFile("student_info.txt", ios::out | ios::trunc);

    if (!outFile.is_open()) {

        cout << "File not found." << endl;
        return 1;

    }

    outFile << "File has been reset." << endl;
    outFile.close();

    ifstream inFile("student_info.txt");

    if (!inFile.is_open()) {

        cout << "File not found." << endl;
        return 1;

    }

    string line;

    cout << "Updated content of student_info.txt:" << endl;

    while (getline(inFile, line)) {

        cout << line << endl;

    }

    inFile.close();

    return 0;`

}
```

Output:

```
Updated content of student_info.txt:
File has been reset.
```

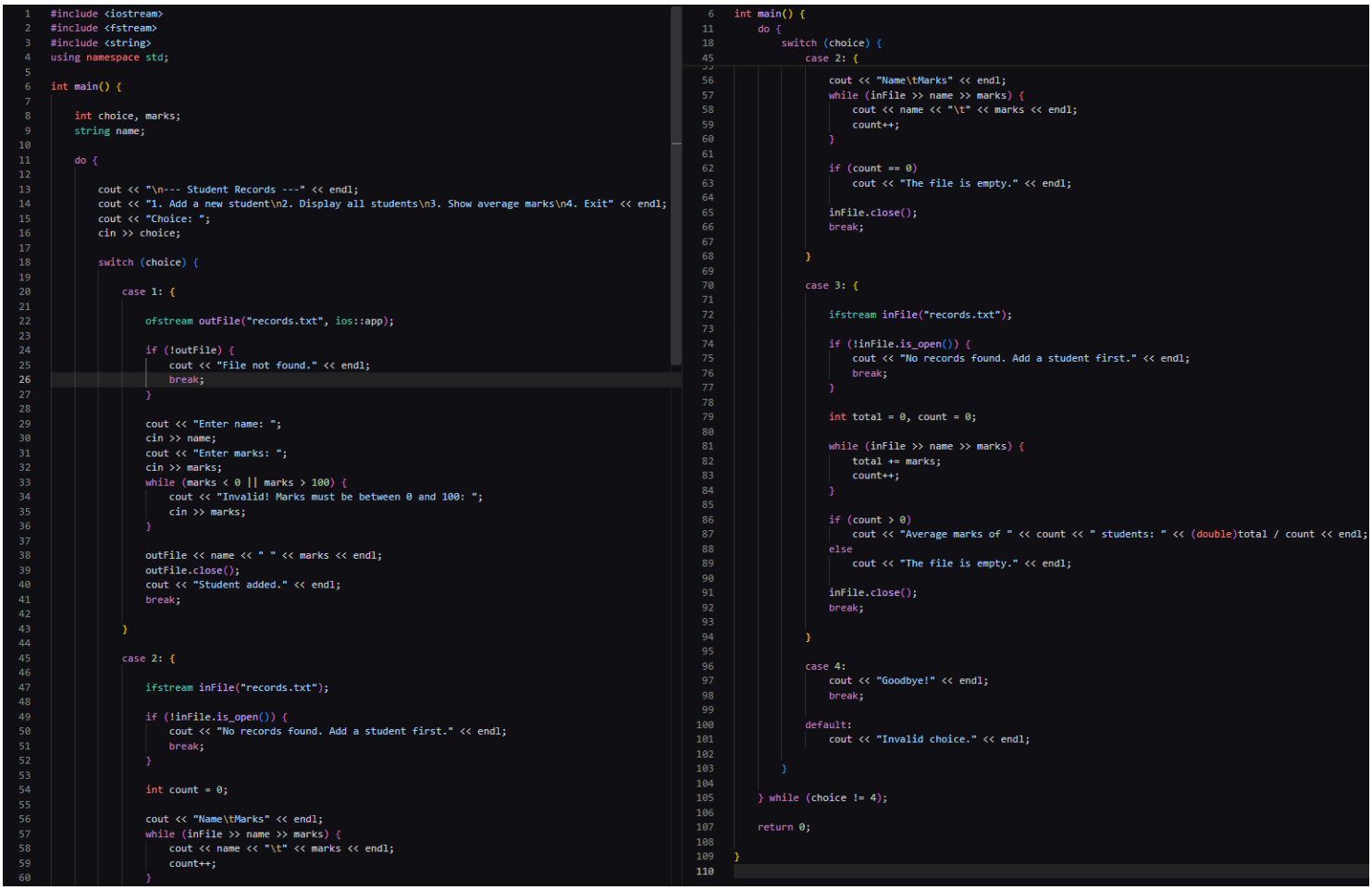
File:

File Edit View

File has been reset.

Home Task: Simple Student Menu

Code (1st half on left, 2nd half on right):



```cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {

    int choice, marks;
    string name;

    do {

        cout << "\n--- Student Records ---" << endl;
        cout << "1. Add a new student\n2. Display all students\n3. Show average marks\n4. Exit" << endl;
        cout << "Choice: ";
        cin >> choice;

        switch (choice) {

            case 1: {

                ofstream outFile("records.txt", ios::app);

                if (!outFile) {
                    cout << "File not found." << endl;
                    break;
                }

                cout << "Enter name: ";
                cin >> name;
                cout << "Enter marks: ";
                cin >> marks;
                while (marks < 0 || marks > 100) {
                    cout << "Invalid! Marks must be between 0 and 100: ";
                    cin >> marks;
                }

                outFile << name << " " << marks << endl;
                outFile.close();
                cout << "Student added." << endl;
                break;

            }

            case 2: {

                ifstream inFile("records.txt");

                if (!inFile.is_open()) {
                    cout << "No records found. Add a student first." << endl;
                    break;
                }

                int count = 0;

                cout << "Name\tMarks" << endl;
                while (inFile >> name >> marks) {
                    cout << name << "\t" << marks << endl;
                    count++;
                }

                if (count == 0)
                    cout << "The file is empty." << endl;

                inFile.close();
                break;

            }

            case 3: {

                ifstream inFile("records.txt");

                if (!inFile.is_open()) {
                    cout << "No records found. Add a student first." << endl;
                    break;
                }

                int total = 0, count = 0;

                while (inFile >> name >> marks) {
                    total += marks;
                    count++;
                }

                if (count > 0)
                    cout << "Average marks of " << count << " students: " << (double)total / count << endl;
                else
                    cout << "The file is empty." << endl;

                inFile.close();
                break;

            }

            case 4:
                cout << "Goodbye!" << endl;
                break;

            default:
                cout << "Invalid choice." << endl;

        }

    } while (choice != 4);

    return 0;

}
```

Output (1st half on left, 2nd half on right):

```
--- Student Records ---
1. Add a new student
2. Display all students
3. Show average marks
4. Exit
Choice: 1
Enter name: aaliyan
Enter marks: 120
Marks must be between 0 and 100: 67
Student added.

--- Student Records ---
1. Add a new student
2. Display all students
3. Show average marks
4. Exit
Choice: 2
Name    Marks
aaliyan 67
```

```
--- Student Records ---
1. Add a new student
2. Display all students
3. Show average marks
4. Exit
Choice: 3
Average marks of 2 students: 67

--- Student Records ---
1. Add a new student
2. Display all students
3. Show average marks
4. Exit
Choice: 4
Goodbye.
```